

Solution Requirements

Functional and non-functional requirements for the documented AI BlogNest API.

Item	Version	Team ID	Document ID
Project Name	aiblognest	Maximum Marks	4 Marks

Step 1: Functional Requirements (FR)

Item	Requirement Category	Requirement Description	Priority
1	Authentication	Users can register and log in using password hashing and JWT-based access.	High
2	Authorization levels	Protected routes require a valid JWT. A separate multi-role model is not specified.	Medium
3	External interfaces	Integrate Google Gemini for AI generation/summarization and MongoDB for persistence.	High
4	Transactions processing	Handle blog create, read, update and delete through REST endpoints.	High
5	Reporting	Provide API responses; a dedicated reporting dashboard/export is not specified.	Low
6	Business rules, etc.	Validate requests, protect authenticated operations and manage blog content through the documented workflow.	High
7	Compliance to laws or regulations	No specific legal/industry compliance requirement is documented.	Low
8	Other	Provide AI blog generation from a topic and AI summarization of supplied content.	High

Step 2: Non-Functional Requirements (NFR)

Item	Requirement Category	Requirement Description	Requirement Status
1	Performance & Speed	API endpoints should respond efficiently under normal load; AI endpoints depend on an external service.	Establish through executed performance testing; no measured result supplied.
2	Scalability	Support growth through API optimization, database scaling and horizontal deployment when required.	Validate with load testing and monitoring; no capacity figure supplied.
3	Security & Data Privacy	Use JWT, bcrypt password hashing and environment-based secrets.	No production security audit supplied; never expose secrets.
4	Reliability & Availability	Handle expected failures and return appropriate errors.	Production uptime target not supplied; validate after deployment.
5	Usability & Accessibility	API requests/responses should be clear and testable through Postman/Thunder Client.	No frontend accessibility requirement specified.
6	Other — Maintainability	Use modular MVC organization, environment configuration and documented API endpoints.	Verify against complete source repository.